Executive Control in Basque–Spanish young and older bilinguals: the role of individual differences across the adult lifespan

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Although bilingualism has long been proposed to enhance executive control (EC) (Bialystok, 2017), evidence for this link remains mixed (Paap, 2019). The proposed bilingual advantage assumes that managing two co-activated languages—through selection, switching, and inhibition (Miyake et al., 2000)—enhances domain-general EC. Yet findings are inconsistent across studies, likely due to theoretical and methodological variability. Recent approaches emphasize bilingualism as an adaptive, experience-driven phenomenon (Bialystok & Craik, 2022), highlighting the need to model both bilingualism and EC as continuous, multifaceted constructs. This study examined how age of acquisition (AoA), proficiency, language use, and switching frequency modulate three EC components—shifting, inhibition, and updating—in younger and older Basque–Spanish bilinguals.

Participants were 124 younger adults (*M* = 21.31, range 18–35) and 98 older adults (*M* = 65.63, range 60–79) from the Basque Country (Spain), representing a wide range of bilingual experiences. We assessed bilingualism through self-report- and objective measures including naming, lexical decision, and Basque knowledge tests for proficiency and a voluntary switching task for language switching. Fluid intelligence (Raven-2) was included as a covariate, and OA cognitive health was screened (MMSE-37). Participants completed six tasks (3 verbal and 3 non-verbal) targeting shifting (Verbal Fluency; Trail Making Test), inhibition (Stroop; Flanker), and updating (Keep Track; N-back). Mixed-effects regressions were applied to combined data from both age groups.

Bilingualism modulated EC in domain-specific ways. Neither bilingualism nor age affected cognitive flexibility, likely reflecting task simplicity and/or lower sensitivity of shifting to bilingual experience. Language use influenced inhibition, with more balanced use reducing Stroop interference similarly across age groups. For updating, frequent language switching improved accuracy in the Keep-Track task, but only under low cognitive load. In the N-back task, language use interacted with fluid intelligence: more balanced language use predicted accuracy more strongly as fluid intelligence increased.

Age impacted inhibition and updating. As expected, older adults showed slower reaction times (RTs) and lower accuracy than young adults, particularly under high cognitive demands. In inhibition, older adults exhibited larger Stroop effects but smaller Flanker effects, suggesting preserved conflict sensitivity. Moreover, only older adults were affected by block-level monitoring demands, slowing in the block with low predictability of incongruent trials, suggesting that trial-level conflict resolution is maintained with age, whereas strategic adaptation to context declines. In updating, OA showed reduced updating efficiency and greater difficulty with conflict monitoring under higher cognitive load.

Overall, these findings show that bilingualism modulates executive control in domain-specific ways, with different bilingual dimensions influencing performance across tasks and age groups. Notably, OA showed comparable performance to YA in demanding updating conditions and reduced conflict effects in inhibition. In these cases, fluid intelligence predicted OA performance, suggesting a potential compensatory reliance on domain-general cognitive resources, which may reflect a form of cognitive reserve shaped by lifelong bilingual experience, helping to sustain executive functioning in older age.

References

Bialystok, E. (2017). The bilingual adaptation: How minds accommodate experience. Psychological Bulletin, 143(3), 233–262.

Paap, K. R. (2019). The bilingual advantage debate. In J. W. Schwieter (Ed.), The handbook of the neuroscience of multilingualism (pp. 701–719).

Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex “frontal lobe” tasks: A latent variable analysis. Cognitive Psychology, 41(1), 49–100.

Bialystok, E., & Craik, F. I. M. (2022). How does bilingualism modify cognitive function? Attention to the mechanism. Psychonomic Bulletin & Review, 29, 2300–2309.

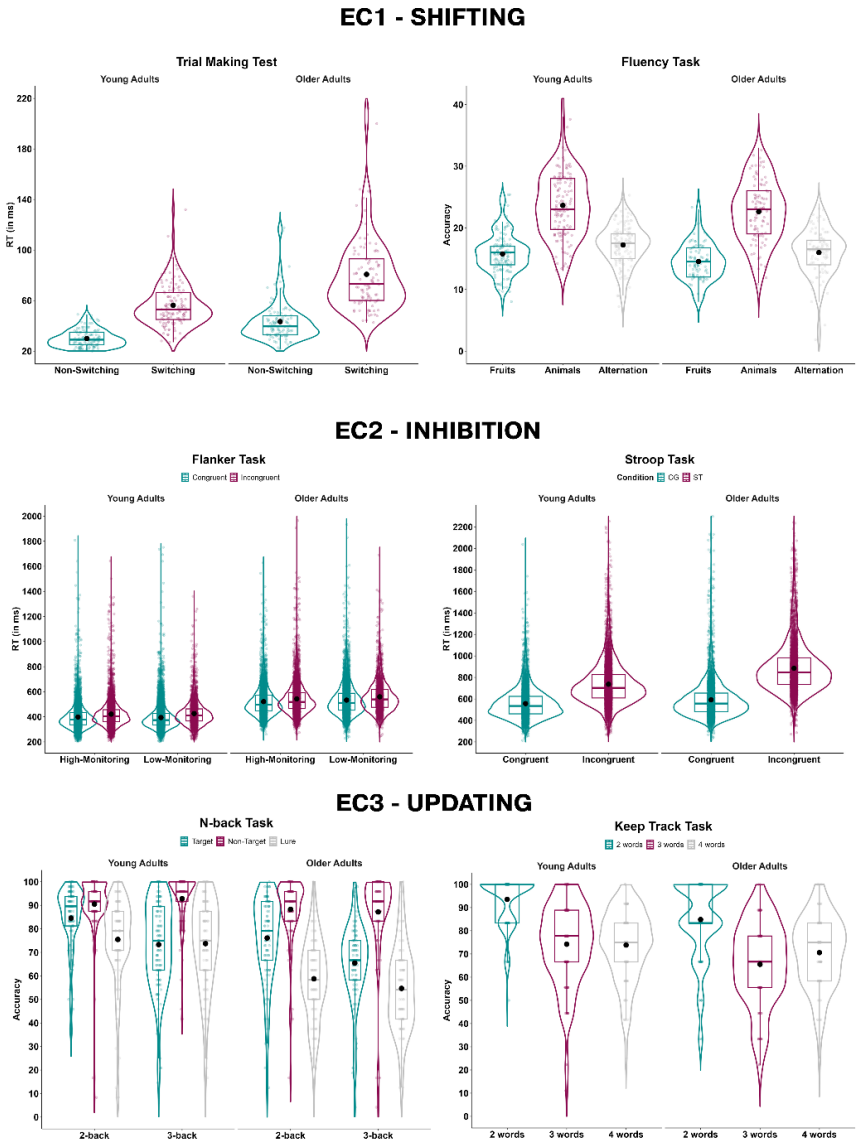

Figure 1. The figure shows raw data from six EC tasks: shifting (top), inhibition (middle), and updating (bottom). Each EC function includes a non-verbal task (left) and a verbal task (right). Young adults are

shown on the left of each plot, and older adults on the right. Violin plots, boxplots, and jittered points display the distribution and variability of individual performance.